MTH636 – Game Theory

Lecture 10: Existence of Equilibria

1 Fundamental Theorems of Game Theory

Two of the most important theorems in game theory guarantee the existence of solutions, provided that we allow players to use mixed strategies and the game is finite.

Theorem: Von Neumann's Minimax Theorem (1928)

Every two-player zero-sum game where both players have **finitely many pure strategies** has a **value** in mixed strategies.
That is, $\underline{V} = \overline{V}$ when players are allowed to randomize.

Theorem: Nash's Existence Theorem (1950)

Every strategic form game where all players have **finitely many pure strategies** has at least one **Nash Equilibrium** in mixed strategies.

Note

These theorems require the set of pure strategies to be **finite**. If the strategy space is infinite (e.g., choosing any real number, or choosing any integer), an equilibrium or value is **not guaranteed** to exist, even in mixed strategies.

2 Counterexample: Infinite Strategy Spaces

Let's see an example where Von Neumann's Minimax Theorem fails because the strategy spaces are infinite.

Example: The Integer Game

$N = \{1, 2\}$. Both players independently choose a natural number: $S_1 = S_2 = \mathbb{N} = \{1, 2, 3, \dots\}$. The player who picks the larger number wins:

$$u_1(s_1, s_2) = \begin{cases} 1 & \text{if } s_1 > s_2 \\ 0 & \text{if } s_1 = s_2 \\ -1 & \text{if } s_1 < s_2 \end{cases}$$

This is a zero-sum game.

2.1 Pure Strategy Value

- **Player 1's Max-Min:** $\sup_{s_1} \inf_{s_2} u_1(s_1, s_2) = -1$. (No matter what s_1 Player 1 chooses, P2 can always pick $s_2 = s_1 + 1$ to make P1 lose).

- **Player 2's Min-Max:** $\inf_{s_2} \sup_{s_1} u_1(s_1, s_2) = 1$.

Since $-1 \neq 1$, there is no value in pure strategies.

2.2 Mixed Strategy Value

Does it have a value in mixed strategies? Let Σ_i be the set of probability distributions over $\mathbb{N}$. Fix any mixed strategy $\sigma_1 \in \Sigma_1$ for Player 1. Since probabilities must sum to 1, most of the probability mass must be concentrated on smaller numbers. For any $\epsilon > 0$, there exists an integer k such that Player 1 plays a number $\leq k$ with probability $> 1 - \epsilon$:

$$\sigma_1(\{1, 2, \ldots, k\}) > 1 - \epsilon$$

If Player 2 simply plays the pure strategy $s_2 = k + 1$, Player 2 wins against all of P1's numbers from 1 to k, and only potentially loses to numbers $> k$.

$$U_1(\sigma_1, s_2 = k + 1) = \underbrace{\sigma_1(\{k + 2, \ldots\})}_{\text{P1 wins}}(1) + \underbrace{\sigma_1(\{1, \ldots, k\})}_{\text{P1 loses}}(-1)$$

Since $\sigma_1(\{1, \ldots, k\}) > 1 - \epsilon$, the probability P1 wins is less than ϵ.

$$U_1(\sigma_1, k + 1) < \epsilon(1) + (1 - \epsilon)(-1) = -1 + 2\epsilon$$

Since this is true for any ϵ, the infimum over all of P2's strategies is ≤ -1.

$$\inf_{\sigma_2 \in \Sigma_2} U_1(\sigma_1, \sigma_2) \leq -1 \quad \Longrightarrow \quad \sup_{\sigma_1} \inf_{\sigma_2} U_1(\sigma_1, \sigma_2) = -1$$

By symmetry for Player 2, $\inf_{\sigma_2} \sup_{\sigma_1} U_1(\sigma_1, \sigma_2) = 1$. Thus, **the value does not exist even in mixed strategies**.

3 Simplifying Minimax for Mixed Strategies

For finite two-player zero-sum games, finding the mixed max-min value involves continuous optimization over the simplex Σ_i.

Theorem: Minimizing over Pure Strategies

When calculating the max-min value in mixed strategies, the inner minimization over all mixed strategies σ_2 can be replaced by a minimization over just the **pure** strategies s_2:

$$\underline{V} = \max_{\sigma_1 \in \Sigma_1} \min_{\sigma_2 \in \Sigma_2} U(\sigma_1, \sigma_2) = \max_{\sigma_1 \in \Sigma_1} \min_{s_2 \in S_2} U(\sigma_1, s_2)$$

Similarly:

$$\overline{V} = \min_{\sigma_2 \in \Sigma_2} \max_{\sigma_1 \in \Sigma_1} U(\sigma_1, \sigma_2) = \min_{\sigma_2 \in \Sigma_2} \max_{s_1 \in S_1} U(s_1, \sigma_2)$$

Why? Because $U(\sigma_1, \sigma_2)$ is a weighted average of $U(\sigma_1, s_2)$ for all pure s_2. A weighted average can never be strictly less than the absolute minimum of its components. So the worst-case response by an opponent will always just be to put 100% probability on the single pure strategy that hurts you the most.